

01NAEX - Lecture 02

Comparing Several Treatment Means, Linear Regression

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Today's Goals

Learning goals:

- ▶ Identify **factors**, **levels**, and **responses**.
- ▶ Formulate hypotheses for a **one-way fixed-effects ANOVA**.
- ▶ Map real problems to the **means/effects** models.
- ▶ State and check key **assumptions** (independence, homoscedasticity, normality).
- ▶ Decide on an appropriate **post-hoc** plan (Tukey/LSD/Dunnett).

N: factors may interact with nuisance conditions (e.g. day).

Good design + correct post-hoc keeps false positives in check.

Plasma etch rate example

- ▶ **Factor (A): Power[W]** with **levels** {160, 180, 200, 220}
 $\Rightarrow a = 4$.
- ▶ **Response (*y*): Etch rate** [Å/min].
- ▶ **Design:** *n* replicates per level (balanced), randomized run order.

Single-Factor Experiment

Suppose we have a **treatment** - different **levels** of a single **factor** that we wish to compare.

There will be, in general, n observations under the i th treatment.

Typical Data for a Single Factor Experiment

Treatment (level)	Observations				Totals	Averages
1	y_{11}	y_{12}	$\cdots$	y_{1n}	$y_{1.}$	$\bar{y}_{1.}$
2	y_{21}	y_{22}	$\cdots$	y_{2n}	$y_{2.}$	$\bar{y}_{2.}$
$\vdots$	$\vdots$	$\vdots$	$\cdots$	$\vdots$	$\vdots$	$\vdots$
a	y_{a1}	y_{a2}	$\cdots$	y_{an}	$y_{a.}$	$\bar{y}_{a.}$
TOTAL					$y_{..}$	$\bar{y}_{..}$

$$y_{i.} = \sum_{j=1}^n y_{ij} \quad y_{..} = \sum_{i=1}^a \sum_{j=1}^n y_{ij} \quad \bar{y}_{i.} = \frac{y_{i.}}{n} \quad \bar{y}_{..} = \frac{y_{..}}{N} \quad i = 1, 2, \dots, a$$

and $N = (a \cdot n)$ is the total number of observations.

In this lesson we will mostly work with the **balanced models**, all factor levels are replicated the same number of times.

Single-Factor Experiment - Models

Means model:

$$y_{ij} = \mu_i + \epsilon_{ij} \quad i = 1, 2, \dots, a \quad j = 1, 2, \dots, n_i,$$

where y_{ij} is the ij th observation, μ_i is the mean of the i th factor level and ϵ_{ij} is random error.

Effects model:

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij} \quad i = 1, 2, \dots, a \quad j = 1, 2, \dots, n_i,$$

where μ is **overall mean** of the measurements and α_i is the i th effect of factor A.

Standard definition of the overall mean is:

$$\mu = \sum_{i=1}^a w_i \mu_i, \quad \text{where } \sum_{i=1}^a w_i = 1,$$

with the most frequent setting: $w_i = \frac{1}{a}$ for $i = 1, 2, \dots, a$.

The weighted average is used when there are an unequal number of observations in each treatment. The weights w_i could be taken as $\frac{n_i}{N}$, for balanced model $n_i = n, \forall i$.

Single-Factor Experiment - Models

Regression model without intercept for a single factor experiment:

$$y_{ij} = \beta_1 x_{1j} + \beta_2 x_{2j} + \cdots + \beta_a x_{aj} + \epsilon_{ij} \quad i = 1, 2, \dots, a \quad j = 1, 2, \dots, n_i,$$

where the regression variables are indicators (i.e. take on the values 1, if observation j is from treatment i , and 0 otherwise)

$$\mathbf{Y} = \begin{pmatrix} \mathbf{y}_{n_1} \\ \mathbf{y}_{n_2} \\ \vdots \\ \mathbf{y}_{n_a} \end{pmatrix} = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{1}_{n_2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{1}_{n_a} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_a \end{pmatrix} + \begin{pmatrix} \epsilon_{n_1} \\ \epsilon_{n_2} \\ \vdots \\ \epsilon_{n_a} \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

The relationship between the parameters in the Means and Regression model:

$$\beta_i = \mu_i \quad i = 1, 2, \dots, a.$$

Single-Factor Experiment - Models

Regression model with intercept for a single factor experiment:

$$y_{ij} = \beta_1 x_{1j} + \beta_2 x_{2j} + \cdots + \beta_a x_{aj} + \epsilon_{ij} \quad i = 1, 2, \dots, a \quad j = 1, 2, \dots, n_i,$$

where $x_{1j} = 1, \forall j$, and the others regression variables are indicators (i.e. take on the values 1, if observation j is from treatment i , and 0 otherwise)

$$\mathbf{Y} = \begin{pmatrix} \mathbf{y}_{n_1} \\ \mathbf{y}_{n_2} \\ \vdots \\ \mathbf{y}_{n_a} \end{pmatrix} = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{1}_{n_2} & \mathbf{1}_{n_2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{1}_{n_a} & \mathbf{0} & \cdots & \mathbf{1}_{n_a} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_a \end{pmatrix} + \begin{pmatrix} \epsilon_{n_1} \\ \epsilon_{n_2} \\ \vdots \\ \epsilon_{n_a} \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

The relationship between the parameters in the Effects and Regression model:

$$\beta_1 = \mu_1 \quad \text{and} \quad \beta_i = \mu_i - \mu_1 \quad i = 2, \dots, a.$$

Analysis of Fixed-Effects Model

We are interested in testing the equality of the a factor levels means:

$$\mathbb{E}(y_{ij}) = \mu + \alpha_i = \mu_i \quad i = 1, 2, \dots, a.$$

The appropriate hypothesis is:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_a.$$

$$H_1 : \mu_i \neq \mu_j \quad \text{for at least one pair } (i, j).$$

If we assume the homoscedasticity (the finite variance σ^2 is constant for all levels of factor and observations are mutually independent) and

$$y_{ij} \sim N(\mu_i, \sigma^2)$$

then the appropriate procedure for testing the equality of several means is:
analysis of variance (ANOVA).

Q: What should we assume? (independence, identical distribution, homoscedasticity, normality, enough observations, ...).

Q: Name some residuals checklist plots.

Decomposition of the Total Sum of Squares

Name **ANOVA** comes from a partitioning of total variability into its components parts.

Overall total variability = between factor levels variability + within variability

$$SS_T = SS_A + SS_E$$

The total corrected sum of squares: $SS_T = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2$

The sum of squares due to treatment A: $SS_A = n \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2$

The error sum of squares: $SS_E = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$
 $= SS_T - SS_A$

Mean squares:

$$MS_A = \frac{SS_A}{a-1}, \quad MS_E = \frac{SS_E}{N-a}.$$

Expected values of the mean squares:

$$\mathbb{E}(MS_E) = \sigma^2, \quad \mathbb{E}(MS_A) = \sigma^2 + \frac{n \sum_{i=1}^a \alpha_i^2}{a-1}.$$

MS_E estimates σ^2 , and, if there are no differences in treatment means, MS_A is also an unbiased estimator of σ^2 .

ANOVA Table for Single-Factor, Fixed-Effect, Balanced Model

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Between treatments	$SS_A = n \sum_{i=1}^a (\bar{y}_i - \bar{y}_{..})^2$	$a - 1$	MS_A	$F_0 = \frac{MS_A}{MS_E}$
Within treatments	$SS_E = SS_T - SS_A$	$N - a$	MS_E	
Total	$SS_T = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2$	$N - 1$		

If data are unbalanced, the manual computational formulas for SS_T and SS_A :

$$SS_T = \sum_{i=1}^a \sum_{j=1}^{n_i} y_{ij}^2 - \frac{y_{..}^2}{N} \quad \text{and} \quad SS_A = \sum_{i=1}^a \frac{y_{i.}^2}{n_i} - \frac{y_{..}^2}{N},$$

where n_i is number of observations taken under treatment i and $N = \sum_{i=1}^a n_i$.

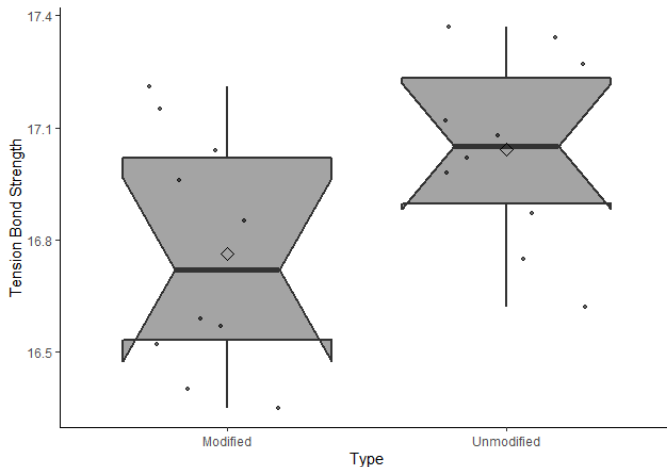
Cochran's theorem implies that $\frac{SS_A}{\sigma^2}$ and $\frac{SS_E}{\sigma^2}$ are independently distributed chi-square random variables and the **test statistic** is given by:

$$F_0 = \frac{\frac{SS_A}{a-1}}{\frac{SS_E}{N-a}} = \frac{MS_A}{MS_E}.$$

ANOVA for a Simple Comparative Experiment

In simple comparative experiment the ANOVA model changes into simple t-test.

Example from lecture 01:



ANOVA for a Simple Comparative Experiment

In a simple comparative experiment (see Lecture 01) the ANOVA model reduces to the two-sample t -test: $F_0 = t_0^2$ and the p -values coincide.

Example from Lecture 01 (mortar data) in Python:

```
stats.ttest_ind(modified, unmodified, equal_var=True)
t = -2.1869, df = 18, p-value = 0.0422
```

```
fit = smf.ols("y ~ C(mortar)", data=df).fit()
sm.stats.anova_lm(fit, typ=2)
```

	sum_sq	df	F	PR(>F)
C(mortar)	0.38642	1.0	4.7824	0.042197
Residual	1.45440	18.0		

$$F_0 = 4.782 = (-2.1869)^2 \text{ and } MS_E = 1.4544/18 = 0.0808 = s_p^2.$$

Example - Plasma Etching Experiment: Data

Etch Rate Data (in A/min) from Plasma Etching Experiment

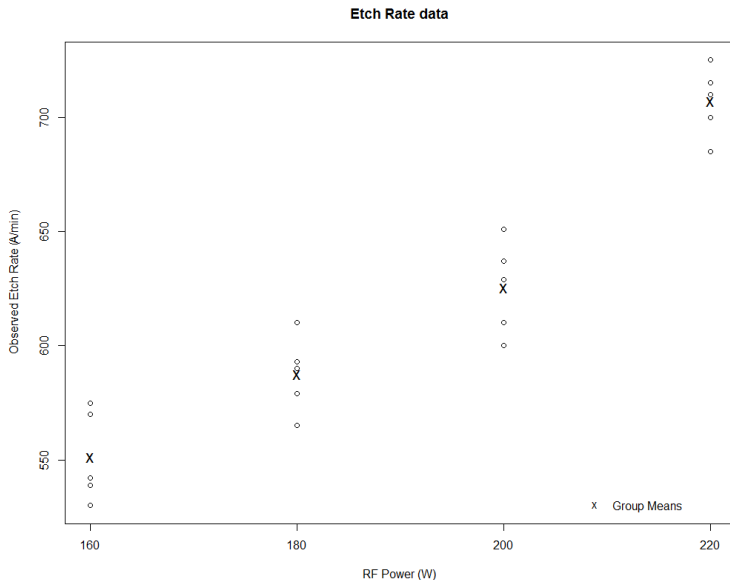
Power (W)	Observations (5 runs)					Totals	Averages
	1	2	3	4	5		
160	575	542	530	539	570	2 756	551.2
180	565	593	590	579	610	2 937	587.4
200	600	651	610	637	629	3 127	625.4
220	725	700	715	685	710	3 535	707.0
						$y_{..} = 12355$	$\bar{y}_{..} = 617.75$

20 runs were made in random order.

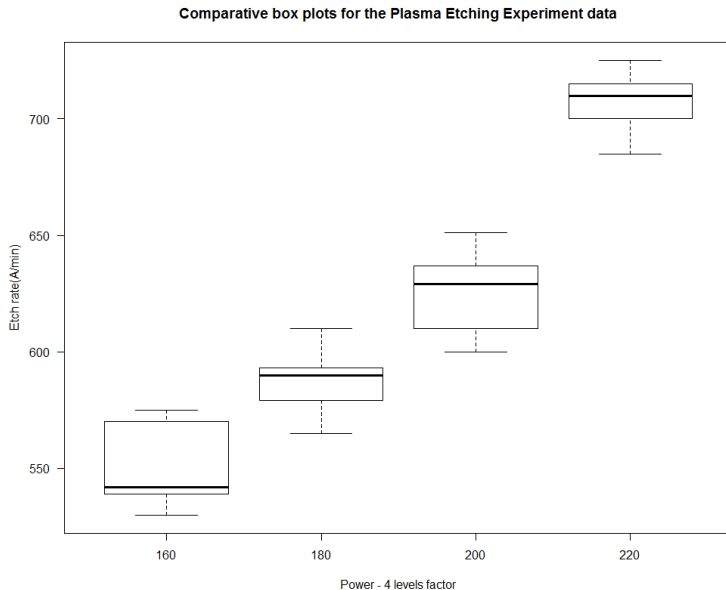
Performing all 6 pairwise t -tests is inefficient. It takes a lot of effort and it inflates the type I error.

Q: Should we block by day or by operator if needed?

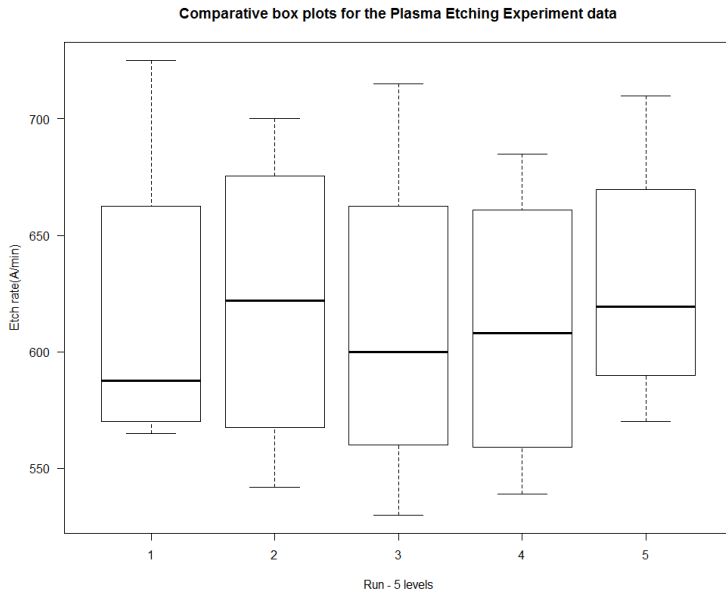
Example - Plasma Etching Experiment: Data



Example - Plasma Etching Experiment: Data



Example - Plasma Etching Experiment: Data



ANOVA for the Plasma Etching Experiment

Two models: power only, and power plus the run index. The run index is not a designed block; the comparison only shows that there is no run-to-run drift.

```
m1 = smf.ols("rate ~ C(Power)", data=etch).fit()
m2 = smf.ols("rate ~ C(Power) + C(Run)", data=etch).fit()
```

```
sm.stats.anova_lm(m2, typ=2)          # power + run
```

	sum_sq	df	F	PR(>F)
C(Power)	66870.55	3.0	62.369	1.37e-07
C(Run)	1050.50	4.0	0.735	0.586
Residual	4288.70	12.0		

```
sm.stats.anova_lm(m1, typ=2)          # power only
```

	sum_sq	df	F	PR(>F)
C(Power)	66870.55	3.0	66.797	2.88e-09
Residual	5339.20	16.0		

```
sm.stats.anova_lm(m1, m2)             # does Run add anything? No.
```

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	16.0	5339.2				
1	12.0	4288.7	4.0	1050.5	0.735	0.5857

ANOVA for the Plasma Etching Experiment

```
m1 = smf.ols("rate ~ C(Power)", data=etch).fit()
sm.stats.anova_lm(m1, typ=2)
```

	sum_sq	df	F	PR(>F)
C(Power)	66870.55	3.0	66.797	2.88e-09
Residual	5339.20	16.0		

$$SS_E = SS_{Residual} = SS_{Total} - SS_{Power} = 72209.75 - 66870.55 = 5339.20$$

$$F_0 = \frac{22290.18}{333.7} = 66.8$$

Because $F_0 = 66.8 > 5.29 = F_{0.01,3,16}$ we reject H_0 at the significance level $\alpha = 0.01$ and conclude that the treatment means differ.

Effects Model for the Plasma Etching Experiment

Treatment coding (statsmodels default): the intercept is $\bar{y}_1$. (160 W), the other coefficients are differences from the 160 W level, i.e. $\beta_i = \mu_i - \mu_1$.

```
m1 = smf.ols("rate ~ C(Power)", data=etch).fit()
```

```
m1.summary().tables[1]
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	551.200	8.169	67.471	0.000	533.882	568.518
C(Power) [T.180]	36.200	11.553	3.133	0.006	11.708	60.692
C(Power) [T.200]	74.200	11.553	6.422	0.000	49.708	98.692
C(Power) [T.220]	155.800	11.553	13.485	0.000	131.308	180.292

```
m1.conf_int(alpha=0.05)      # the same 95% confidence intervals
```

Residuals: min -25.4, median 2.8, max 25.6; $\hat{\sigma} = \sqrt{MS_E} = \sqrt{333.7} = 18.27$
on 16 df.

Means Model for the Plasma Etching Experiment

Without the intercept the coefficients are the treatment means $\hat{\mu}_i = \bar{y}_{i.}$, all with the same standard error $\sqrt{MS_E/n} = 8.169$.

```
m0 = smf.ols("rate ~ C(Power) - 1", data=etch).fit()  
m0.summary().tables[1]
```

	coef	std err	t	P> t	[0.025	0.975]
C(Power) [160]	551.200	8.169	67.471	0.000	533.882	568.518
C(Power) [180]	587.400	8.169	71.902	0.000	570.082	604.718
C(Power) [200]	625.400	8.169	76.553	0.000	608.082	642.718
C(Power) [220]	707.000	8.169	86.542	0.000	689.682	724.318

Means Model for the Plasma Etching Experiment

100(1 - α) percent confidence interval on the i th treatment mean μ_i is:

$$\bar{y}_{i.} - t_{\frac{\alpha}{2}, N-a} \sqrt{\frac{MS_E}{n}} \leq \mu_i \leq \bar{y}_{i.} + t_{\frac{\alpha}{2}, N-a} \sqrt{\frac{MS_E}{n}}$$

```
m0.conf_int(alpha=0.05)
```

	lower	estimate	upper
C(Power) [160]	533.882	551.200	568.518
C(Power) [180]	570.082	587.400	604.718
C(Power) [200]	608.082	625.400	642.718
C(Power) [220]	689.682	707.000	724.318

Example of the 95% confidence interval for treatment 4:

$$689.68 = 707 - 2.120 \sqrt{\frac{333.7}{5}} \leq \mu_4 \leq 707 + 2.120 \sqrt{\frac{333.7}{5}} = 724.32$$

Treatment Effects in the Plasma Etching Experiment

```
grand_mean = etch["rate"].mean() # 617.75
means = etch.groupby("Power")["rate"].mean()
effects = means - grand_mean # tau_i = ybar_i. - ybar..
      160      180      200      220
-66.55  -30.35      7.65   89.25

MSE = ml.mse_resid # 333.7
np.sqrt(MSE / 5) # SE of a treatment mean 8.169
np.sqrt(MSE * (1/5 - 1/20)) # SE of an effect tau_i 7.075
np.sqrt(MSE / 20) # SE of the grand mean 4.085
```

The effects sum to zero. $\text{Var}(\hat{\tau}_i) = \sigma^2(1/n - 1/N)$, smaller than σ^2/n , because $\bar{y}_{i.}$ and $\bar{y}_{..}$ are positively correlated.

ANOVA - Model Adequacy Checking

Check the three pillars:

- ▶ **Independence** (no time/sequence trends)
- ▶ **Constant variance** (homoscedasticity across levels)
- ▶ **Normal errors** (for inference on means)

Residuals and scalings (one-way fixed effects):

$$e_{ij} = y_{ij} - \hat{y}_{ij} = y_{ij} - \bar{y}_{i\cdot}, \quad \text{with leverage } h_{ij} = 1/n_i.$$

$$\text{Studentized residuals: } r_{ij} = \frac{e_{ij}}{s \sqrt{1 - h_{ij}}}, \quad s = \sqrt{MS_E}.$$

Why studentize? It properly scales by the local variance of e_{ij} .

Core diagnostic plots:

- ▶ r_{ij} vs **fitted** $\hat{y}_{ij}$ (or group means)
- ▶ r_{ij} vs **factor level** (to see level-wise spread)
- ▶ r_{ij} vs **run/order** (independence/drift)
- ▶ **Normal QQ-plot** of residuals (normality)

Residual Patterns - What They Suggest

Residuals vs Fitted:

- ▶ *Trumpet* shape indicates variance increases with mean
⇒ try **variance-stabilizing transform**.
- ▶ *Curvature/arch* (systematic bend) indicates **non-additivity**/scale issues
⇒ try a transform or examine hidden factors, second order term.
- ▶ *Random band* with constant spread ⇒ consistent with homoscedasticity.

r vs Factor level:

- ▶ Level-wise spread differs ⇒ **heteroscedasticity** across levels.
- ▶ Centered, similar spreads ⇒ supports constant variance.

r vs Run/Order (time):

- ▶ Trend, cycles, or blocks ⇒ **dependence/drift**
consider blocking by time, day or adding an order covariate.
- ▶ Structureless cloud ⇒ supports independence.

Normal QQ-plot:

- ▶ Mild S-shape acceptable for small n
severe tails/outliers ⇒ consider robust/transform.

When Diagnostics Fail - What to Do

Heteroscedasticity (non-constant variance):

- ▶ Consider **Brown–Forsythe/Levene** tests for variance checks.
- ▶ **Transform** response (Box–Cox, log, square-root) and re-check.
- ▶ ANOVA is quite robust, but in some cases **Welch's ANOVA** or Model variance (WLS) can be used.

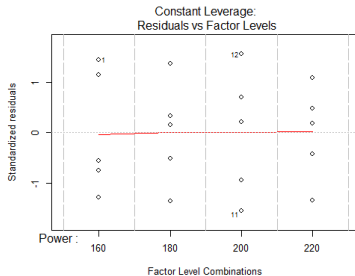
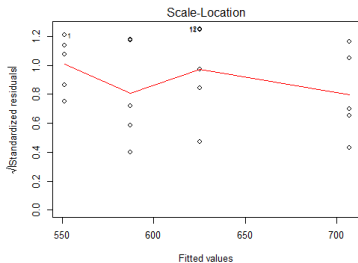
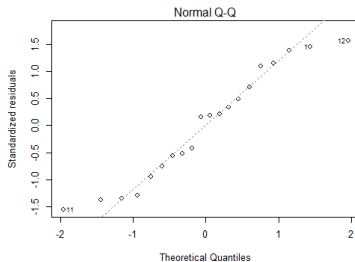
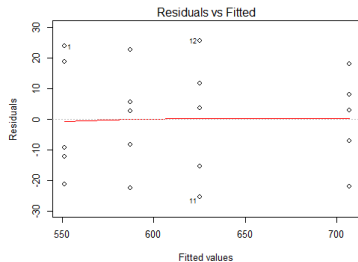
Non-normal errors (esp. small n or strong tails):

- ▶ Transform (as above) and re-check QQ.
- ▶ **Kruskal–Wallis** (rank-based) for location shift across groups.
- ▶ Try to get more data points.

Dependence / run-order effects:

- ▶ **Randomize** run order.
- ▶ **block** by time/day/operator or use mixed-effects.

Model Adequacy Checking for the Plasma Etching Experiment



Model Adequacy Checking for the Plasma Etching Experiment - Equality of Variance

Bartlett's test (likelihood-ratio type, very sensitive to non-normality):

```
groups = [g["rate"].to_numpy(float)
           for _, g in etch.groupby("Power")]
stats.bartlett(*groups)
K^2 = 2.681, df = 3, p-value = 0.443
```

When normality is doubtful use **Levene's test** in the Brown–Forsythe form: the ANOVA F statistic applied to the absolute deviations from the group medians.

```
stats.levene(*groups, center="median")
F = 0.196, df = (3, 16), p-value = 0.898
```

Neither test gives evidence against equal variances; the Brown–Forsythe result is the one to report.

Model Adequacy Checking for the Plasma Etching Experiment - Normality

Normality is an assumption about the errors, so test the **residuals** of the fitted model (20 values), not a single level with $n = 5$.

```
resid = m1.resid
stats.shapiro(resid)                                # Shapiro-Wilk
    W = 0.9375, p-value = 0.215

z = (resid - resid.mean()) / resid.std(ddof=1)
stats.kstest(z, "norm")                             # Kolmogorov-Smirnov
    D = 0.112, p-value = 0.940

[stats.shapiro(g).pvalue for g in groups]           # per level, n = 5
    [0.276, 0.973, 0.870, 0.940]
```

Together with the Q-Q plot: no evidence against normality. With $n = 5$ per level the per-level tests have almost no power, so they prove little either way.

The Plasma Etching Experiment - Regression Model

The factor Power is quantitative, so we can also fit a regression in the actual power values (linear and quadratic):

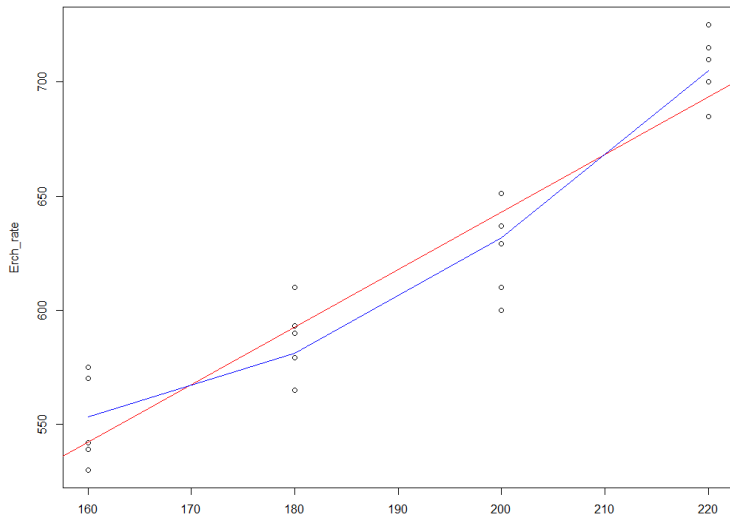
```
r1 = smf.ols("rate ~ RF", data=etch).fit()
      coef    std err          t      P>|t|
Intercept  137.620    41.211     3.339    0.004
RF          2.527     0.215    11.731    0.000
R^2 = 0.884, adj. R^2 = 0.878, s = 21.54 on 18 df

r2 = smf.ols("rate ~ RF + I(RF**2)", data=etch).fit()
      coef    std err          t      P>|t|
Intercept 1147.770   368.521     3.115    0.006
RF         -8.256     3.920    -2.106    0.050
I (RF ** 2)  0.028     0.010     2.754    0.014
R^2 = 0.920, adj. R^2 = 0.911, s = 18.43 on 17 df

sm.stats.anova_lm(r1, r2)  # F = 7.58, p = 0.014: curvature
```

The Plasma Etching Experiment - Regression Model

Comparison of two regressions models, with and without curvature (square of explanatory variable).



Multiple Comparisons (Multiple Testing) - Concept

Why it matters (pairwise means): With a groups, the number of pairwise mean comparisons is

$$m = \frac{a(a-1)}{2}.$$

Testing each at $\alpha = 0.05$ inflates the chance of at least one false positive (**FWER**).

Under the global null (all H_0 true, independent tests):

$$\mathbb{P}(\text{at least one FP}) = 1 - (1 - \alpha)^m, \quad \mathbb{E}[\#\text{FP}] = m\alpha.$$

Example: $m = 100$ tests, $\alpha = 0.05 \Rightarrow 1 - 0.95^{100} \approx 0.994$; expected FP = 5.

Two control goals

- ▶ **FWER** (family-wise error rate): $\mathbb{P}(\geq 1 \text{ FP})$ - conservative approach
 $\Rightarrow$ try Tukey, Bonferroni/Holm/Hochberg.
- ▶ **FDR** (false discovery rate): $\mathbb{E} \left[\frac{\#\text{FP}}{\#\text{Discoveries} \vee 1} \right]$ - less stringent approach
 $\Rightarrow$ try Benjamini–Hochberg, Benjamini–Yekutieli.

Sketch of the timeline

- ▶ Tukey (1953): Honestly Significant Difference **HSD** for all-pairs means (studentized range).
- ▶ Scheffé (1953): simultaneous Confidence Intervals for **all contrasts** (conservative approach).
- ▶ Holm (1979): **Holm–Bonferroni** step-down FWER control (more powerful than Bonferroni).
- ▶ Hochberg (1988): step-up FWER control (slightly stronger than Holm, needs independence or positive dependence).
- ▶ Benjamini–Hochberg (1995): **BH** FDR control.

Multiple Comparisons in Python (Pairwise Means)

Python (statsmodels / scipy):

```
import statsmodels.api as sm
import statsmodels.formula.api as smf
from statsmodels.stats.multicomp import MultiComparison
from statsmodels.stats.multitest import multipletests
from scipy import stats

# ANOVA
fit = smf.ols("y ~ C(group)", data=df).fit()
anova_table = sm.stats.anova_lm(fit, typ=2)

# Tukey HSD (all pairs, FWER control)
mc = MultiComparison(df["y"], df["group"])
tukey = mc.tukeyhsd(alpha=0.05)
print(tukey); tukey.plot_simultaneous()

# pairwise t-tests (pooled MSE), then adjust the  $\alpha/2$  p-values
rej, p_adj, _, _ = multipletests(pvals, alpha=0.05, method="holm")
# method = "bonferroni" | "holm" | "fdr_bh" (FWER, FWER, FDR)

# Dunnett: every treatment against one control
stats.dunnett(y_180, y_200, y_220, control=y_160)
```

Dunnett's Test and FDR for the Plasma Etching Experiment

If 160 W is the current (control) setting, Dunnett's test compares every other level with it only: $a - 1 = 3$ comparisons, hence more power than all-pairs Tukey.

```
stats.dunnett(y180, y200, y220, control=y160)
```

comparison	statistic	p-value	95% CI
180 - 160	3.133	0.017	(6.24, 66.16)
200 - 160	6.422	0.000	(44.24, 104.16)
220 - 160	13.485	0.000	(125.84, 185.76)

For many comparisons the FDR (Benjamini–Hochberg) is the less stringent alternative to FWER control:

```
multipletests(pvals, method="fdr_bh")[1] # 6 raw p-values
```

180-160	200-160	220-160	200-180	220-180	220-200
6.4e-03	1.3e-05	2.2e-09	5.6e-03	5.1e-08	5.4e-06

FWER Control for Pairwise Mean Comparisons (Post-hoc)

Bonferroni (simple, conservative)

Test each at α/m or use $p_{\text{adj}} = \min(mp, 1)$.

N: quick, small m , arbitrary dependence.

Holm-Bonferroni (step-down)

1. Sort $p_{(1)} \leq \dots \leq p_{(m)}$.
2. For $k = 1 \dots m$, compare $p_{(k)}$ to $\alpha/(m - k + 1)$ until first fail, reject all before it.

N: default FWER choice if not using Tukey; allows any dependence.

Hochberg (step-up)

1. Sort $p_{(1)} \leq \dots \leq p_{(m)}$.
2. For $k = m \dots 1$, if $p_{(k)} \leq \alpha/(m - k + 1)$, reject $p_{(1)}, \dots, p_{(k)}$ and stop.

Tukey HSD (all-pairs means via studentized range q)

N: confirmatory all-pairs after ANOVA; balanced/equal-variance ideal.

Dunnett (vs control only, more power than Tukey for this target)

N: one control vs many treatments.

Post-ANOVA Comparison of Means for the Plasma Etching Experiment

All $m = 6$ pairwise t -tests with the pooled MS_E (16 df): raw and adjusted p -values.

pair	diff	raw p	bonferroni	holm	fdr_bh
180-160	36.2	6.4e-03	3.9e-02	9.3e-03	6.4e-03
200-160	74.2	8.4e-06	5.1e-05	2.5e-05	1.3e-05
220-160	155.8	3.7e-10	2.2e-09	2.2e-09	2.2e-09
200-180	38.0	4.6e-03	2.8e-02	9.3e-03	5.6e-03
220-180	119.6	1.7e-08	1.0e-07	8.5e-08	5.1e-08
220-200	81.6	2.7e-06	1.6e-05	1.1e-05	5.4e-06

```
rej, p_adj, _, _ = multipletests(pvals, alpha=0.05, method="holm")
```

All six differences stay significant at $\alpha = 0.05$ under every adjustment. Bonferroni is the most conservative, Holm is uniformly better and needs no independence, BH controls only the FDR.

Tukey HSD for the Plasma Etching Experiment

Tukey (1953) proposed a procedure for testing hypothesis for which the overall family-wise significance level (FWER) is exactly α when sample sizes are n_i and n_j .

Studentized range statistic:

$$q = \frac{\bar{y}_{max} - \bar{y}_{min}}{\sqrt{\frac{MS_E}{2} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}}$$

where $\bar{y}_{max}$ and $\bar{y}_{min}$ are the largest and smallest sample means.

Tukey's test declares two means significantly different if the absolute value of their sample differences exceeds

$$T_\alpha = q_\alpha(a, f) \sqrt{\frac{MS_E}{2} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)},$$

where a is number of levels and f is number of degrees of freedom for error. $q_\alpha(a, f)$'s are tabularized.

Plasma etching experiment example:

$$T_{0.05} = q_{0.05}(4, 16) \sqrt{\frac{MS_E}{n}} = 4.05 \sqrt{\frac{333.7}{5}} = 33.09$$

Thus, any pair of treatment averages that differ in absolute value by more than 33.09 would imply that the corresponding pair of population means are significantly different.

Tukey HSD for the Plasma Etching Experiment

Simultaneous 95% confidence intervals for all pairwise differences (Tukey's Honest Significant Difference):

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd
tukey = pairwise_tukeyhsd(etch["rate"], etch["Power"], alpha=0.05)
print(tukey)
```

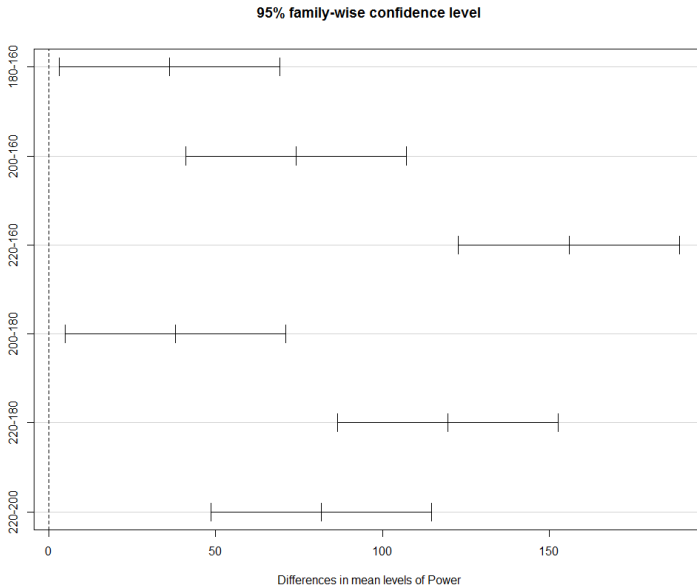
```
Multiple Comparison of Means - Tukey HSD, FWER=0.05
```

group1	group2	meandiff	p-adj	lower	upper	reject
160	180	36.2	0.0294	3.1456	69.2544	True
160	200	74.2	0.0000	41.1456	107.2544	True
160	220	155.8	0.0000	122.7456	188.8544	True
180	200	38.0	0.0216	4.9456	71.0544	True
180	220	119.6	0.0000	86.5456	152.6544	True
200	220	81.6	0.0000	48.5456	114.6544	True

```
tukey.plot_simultaneous() # the plot on the next slide
```

Half-width of every interval = $T_{0.05} = 33.05$ with $q_{0.05}(4, 16) = 4.046$.

Tukey HSD for the Plasma Etching Experiment



Fisher's LSD for the Plasma Etching Experiment

Fisher's Least Significant Difference Method uses the t statistic for testing $H_0 : \mu_i = \mu_j$:

$$t_0 = \frac{\bar{y}_{i.} - \bar{y}_{j.}}{\sqrt{MS_E(\frac{1}{n_i} + \frac{1}{n_j})}}$$

Assuming a two-sided alternative. The quantity

$$LSD = t_{\frac{\alpha}{2}, N-a} \sqrt{MS_E(\frac{1}{n_i} + \frac{1}{n_j})}$$

is called the least significance difference. If $|\bar{y}_{i.} - \bar{y}_{j.}| > LSD$, we conclude that the population means μ_i and μ_j differ.

Fisher's LSD for the Plasma Etching Experiment

Fisher's Least Significant Difference Method uses the t statistic for testing $H_0 : \mu_i = \mu_j$:

$$t_0 = \frac{\bar{y}_i - \bar{y}_j}{\sqrt{MS_E(\frac{1}{n_i} + \frac{1}{n_j})}}$$

Assuming a two-sided alternative. The quantity

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is called the least significance difference. If $|\bar{y}_i - \bar{y}_j| > LSD$, we conclude that the population means μ_i and μ_j differ.

Fisher's LSD Pros

- ▶ High **power** for detecting differences.
- ▶ Simple to explain and compute; uses the common MS_E .

Fisher's LSD Cons

- ▶ **Weak FWER control**: even with a significant ANOVA, FWER can exceed α .
- ▶ Gets riskier as the number of groups a grows (pairs $m = a(a-1)/2$).

Fisher's LSD for the Plasma Etching Experiment

Fisher's Least Significant Difference Method at $\alpha = 0.05$ for the Plasma Etching Experiment is:

$$LSD = t_{0.025, 16} \sqrt{\frac{2MS_E}{n}} = 2.120 \sqrt{\frac{2(333.7)}{5}} = 24.49$$

Thus, any pair of treatment averages that differ in absolute value by more than 24.49 would imply that the corresponding pair of population means are significantly different.

Fisher's LSD for the Plasma Etching Experiment in Python

```
MSE, df_e, n = m1.mse_resid, m1.df_resid, 5      # 333.7, 16, 5
LSD = stats.t.ppf(0.975, df_e) * np.sqrt(2 * MSE / n)
#      = 2.120 * 11.55 = 24.49
```

```
means (Angstrom/min) with 95% CI  ybar_i +- t * sqrt(MSE/n):
160    551.2    (533.9, 568.5)
180    587.4    (570.1, 604.7)
200    625.4    (608.1, 642.7)
220    707.0    (689.7, 724.3)
```

```
pairwise differences against LSD = 24.49:
180-160    36.2    *          200-180    38.0    *
200-160    74.2    *          220-180   119.6    *
220-160   155.8    *          220-200    81.6    *
```

All four means differ from each other (groups a, b, c, d), the same conclusion as Tukey. LSD does not control the FWER, though: with six comparisons and all null hypotheses true, the chance of at least one false positive would be about $1 - (1 - 0.05)^6 \approx 0.26$.

Sample Size Determination

- ▶ Sample size depends on type of experiment, how it will be conducted, resources, and desired sensitivity
- ▶ Sensitivity refers to the difference in means that the experimenter wishes to detect
- ▶ Generally, increasing the number of replications increases the sensitivity or it makes it easier to detect small differences in means

Sample Size Determination

Operating characteristic (OC) curve: For a fixed design, an OC curve plots the type II error β (or power $1 - \beta$) of a test versus a parameter measuring how far the truth is from H_0 .

For single factor fixed effects model:

$$\beta = 1 - \mathbb{P}[\text{Reject } H_0 \mid H_0 \text{ is false}] = 1 - \mathbb{P}[F_0 > F_{\alpha, a-1, N-a} \mid H_0 \text{ is false}]$$

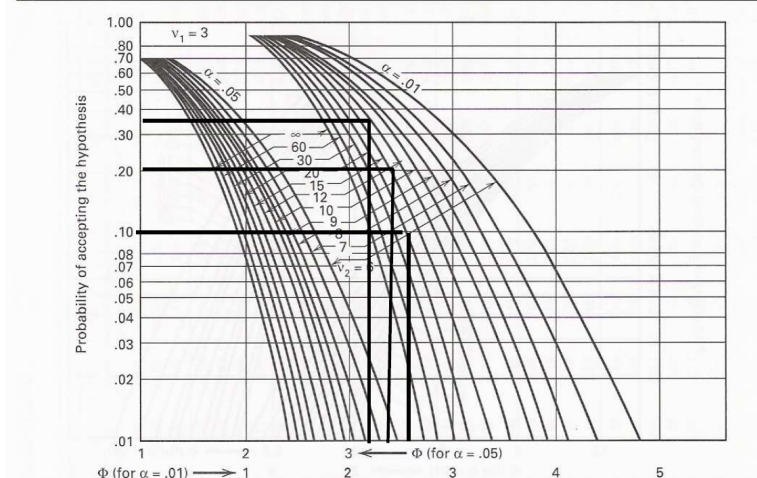
It can be shown that, if H_0 is false, the statistic $F_0 = \frac{MS_A}{MS_E}$ is distributed as a noncentral F random variable with $a - 1$ and $N - a$ degrees of freedom and the noncentrality parameter δ .

Operating characteristic curves plot β against a parameter Φ^2

$$\Phi^2 = \frac{n \sum_{i=1}^a (\mu_i - \bar{\mu})^2}{a\sigma^2}.$$

Sample Size Determination

V. Operating Characteristic Curves for the Fixed Effects Model Analysis of Variance (*continued*)



Sample Size Determination - OC Curves & Noncentral F

One-way fixed-effects ANOVA (balanced: n per group, a groups):

$$F_0 = \frac{MS_A}{MS_E}, \quad H_0 : \mu_1 = \cdots = \mu_a.$$

Under H_1 ,

$$F_0 \sim F_{a-1, N-a}(\lambda), \quad N = an, \quad \lambda = \frac{n}{\sigma^2} \sum_{i=1}^a (\mu_i - \bar{\mu})^2.$$

Convenient scaled distance:

$$\Phi^2 = \frac{1}{a} \frac{n}{\sigma^2} \sum_{i=1}^a (\mu_i - \bar{\mu})^2 \iff \lambda = a\Phi^2.$$

Reading the OC curve: For a chosen α , df $(a-1, N-a)$, and Φ^2 , power is

$$1 - \beta = \mathbb{P}(F_{a-1, N-a}(\lambda) > F_{\alpha; a-1, N-a}), \quad \lambda = a\Phi^2.$$

Deriving the formula $\Phi^2 = \frac{nD^2}{2a\sigma^2}$

Goal (power target): Choose n so that the ANOVA rejects with high probability if *at least one pair of means* differs by D or more.

Set-up: Define

$$S = \sum_{i=1}^a (\mu_i - \bar{\mu})^2, \quad \Phi^2 = \frac{n}{a\sigma^2} S.$$

We want the smallest S among all means satisfying $\max_{i,k} |\mu_i - \mu_k| \geq D$

Minimal S under the constraint: The minimum occurs when (up to symmetry):

$$\mu_1 = +\frac{D}{2}, \quad \mu_2 = -\frac{D}{2}, \quad \mu_3 = \cdots = \mu_a = 0 \quad \Rightarrow \quad \bar{\mu} = 0.$$

Then

$$S = \left(\frac{D}{2}\right)^2 + \left(-\frac{D}{2}\right)^2 + 0 + \cdots + 0 = \frac{D^2}{2}.$$

Plug into Φ^2 :

$$\boxed{\Phi^2 = \frac{n}{a\sigma^2} \cdot \frac{D^2}{2} = \frac{nD^2}{2a\sigma^2}} \quad \Rightarrow \quad \lambda = a\Phi^2 = \frac{nD^2}{2\sigma^2}.$$

From D and MS_E to n - Practical Recipe

Inputs

- ▶ Minimally important difference D between any two means.
- ▶ Pilot (or prior) variance estimate $\hat{\sigma}^2 \approx MS_E$.
- ▶ Number of groups a , significance α , target power $1 - \beta$.

Steps

1. For each trial n , set $N = an$ and df $(a - 1, N - a)$.
2. Choose Φ^2 via the OC curve (or compute power using $\lambda = a\Phi^2$).
3. Use the design constraint $\Phi^2 = \frac{nD^2}{2a\hat{\sigma}^2}$.
4. Rearrange for n :

$$\boxed{n = \frac{2a\hat{\sigma}^2}{D^2} \Phi^2} \quad (\text{pick the smallest integer } n \text{ hitting the desired power}).$$

Q: If the pilot MS_E halves (better measurement), how does n change for the same D and power?

Sample Size Determination for the Plasma Etching Experiment

Goal: Detect any pairwise difference $D = 75$ A/min with power ≥ 0.90 at $\alpha = 0.01$, with $a = 4$ power settings.

Plan:

1. Use pilot $\hat{\sigma}^2 = MS_E$ from a small pre-study (or historical runs).
2. For a candidate n , compute $\Phi^2 = \frac{nD^2}{2a\hat{\sigma}^2}$, so $\lambda = a\Phi^2$.
3. With df $(a-1, an-a)$, compute power $1 - \beta = \mathbb{P}(F_{a-1, an-a}(\lambda) > F_\alpha)$.
4. Increase/decrease n until power ≥ 0.90 .

Rule of thumb: n scales linearly with $\hat{\sigma}^2$ and inversely with D^2 .

$\Rightarrow$ halving σ or doubling D roughly quarters the required n (halving σ^2 halves it).

Sample Size Determination for the Plasma Etching Experiment

In Plasma Etching Experiment, suppose we would like to reject the null hypothesis with a probability of at least 0.90 if any two treatment means differed by as much as 75 A/minute and $\alpha = 0.01$.

Sample Size Determination for the Plasma Etching Experiment

Sample size	ϕ^2	ϕ	DF for errors	Power
4	4.50	2.12	12	0.61
5	5.62	2.37	16	0.80
6	6.75	2.60	20	0.92

Sample Size Determination for the Plasma Etching Experiment

In the Plasma Etching Experiment, suppose we would like to reject the null hypothesis with probability at least 0.90 if any two treatment means differ by as much as $D = 75 \text{ Å/min}$, with $\sigma \approx 25$ and $\alpha = 0.01$.

```
from statsmodels.stats.power import FTestAnovaPower
D, sigma, a, alpha = 75, 25, 4, 0.01
f = (D / sigma) * np.sqrt(1 / (2 * a))      # Cohen's f = 1.061
pw = FTestAnovaPower()
for n in range(4, 11):
    print(n, pw.power(f, nobs=a * n, alpha=alpha, k_groups=a))
n:         4         5         6         7         8         9        10
power:  0.6065  0.8048  0.9154  0.9670  0.9882  0.9961  0.9988

pw.solve_power(f, alpha=alpha, power=0.9, k_groups=a) / a
5.81  ->  n = 6 replicates per level
```

$\Phi^2 = nD^2/(2a\sigma^2) = 1.125n$ and $\lambda = a\Phi^2 = anf^2$; the table on the previous slide is reproduced exactly.

Sample Size Determination for the Plasma Etching Experiment

Power for a given set of means. Here the fitted means are used, so this is a post-hoc power: it only restates how far the data are from H_0 .

```
means = [551.2, 587.4, 625.4, 707.0]; MSE = 333.7; n = 5; a = 4
between_var = np.var(means, ddof=1)    # 4458
f = np.sqrt(between_var * (a - 1) / a / MSE)    # 3.17
pw.power(f, nobs=a * n, alpha=0.01, k_groups=a)    # 1.0000
```

Note that $MS_A = n \cdot \text{between_var} = 22290$. Plugging MS_A in as the variance of the means would overstate the effect n -fold.

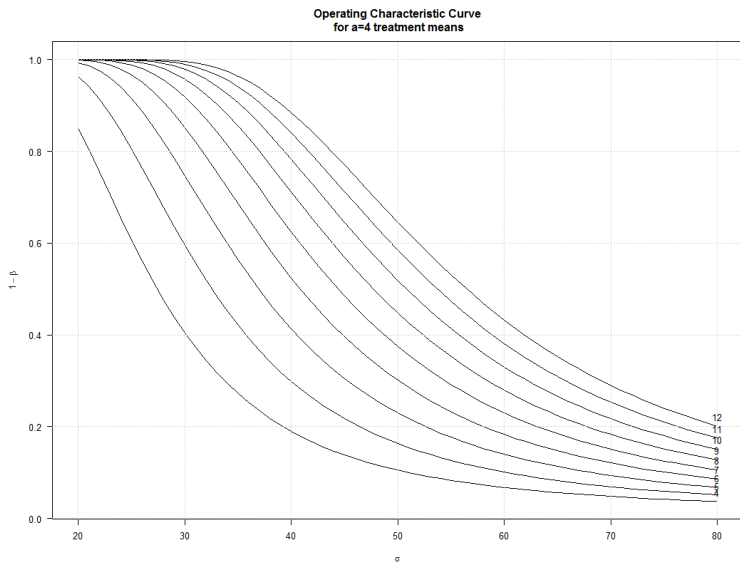
Sample Size Determination

Example of a sample size computation: $a = 4$ groups, variance of the group means 1000, error variance 500, $\alpha = 0.01$, power 0.9.

```
between_var, within_var, a = 1000, 500, 4
f = np.sqrt(between_var * (a - 1) / a / within_var)      # 1.225
pw.solve_power(f, alpha=0.01, power=0.9, k_groups=a) / a
4.76  ->  n = 5 replicates per group
```

(The same $n = 4.76$ as R's `power.anova.test` with `between.var = 1000`, `within.var = 500`.)

Sample Size Determination



Today Exercise & Next Lecture

Today Exercise (Ex02):

- ▶ Problems 3.7–3.9, 3.10 and 3.18–3.19 from Montgomery, Chapter 3.
- ▶ Use the prepared Jupyter notebook code/01NAEX_Ex02.ipynb (Python, Google Colab).
- ▶ Start in class, finish at home; submit by pull request into `HW/` by Sunday 11 October 2026.

Next Lecture: Factorial Designs

- ▶ Main effects and interactions, the two-factor factorial design.
- ▶ Fixed-effects model, ANOVA for two factors, model adequacy checking.
- ▶ General factorial designs. Team Project 1 will be assigned in week 4 (19 October).